Build all 102 forms with a script

DPP Weekly Checks in Google Forms · setup takes about 30 minutes once, then the script builds for about an hour on its own

What the script does. In your **personal** Google Drive it builds every Check, Re-check and Comparison form: the title and description, the Teacher and Period questions, each question's picture, the answer choices, every answer key and point value, the typing rule on typed answers, and the Level question on checks. It also makes a links sheet with each form's status, its edit link and a **copy link** for your school account.
What you still do each week: open the copy link in your school account, publish the copy and post its link (about 2 minutes a form). Students only ever use the school copies, so their responses stay in the school account.

What's in the zip

Item	What it is
DPP Forms kit (folder)	`forms_manifest.json` (every form, question, choice and answer key) and an `images` folder (374 question and level-scale pictures). You upload this folder as is.
Code.gs.txt	The script. You paste it into Apps Script.
appsscript.json.txt	The script's settings: the permissions it asks for.

Setup (once)

1 Upload the kit to your personal Drive · 3 minutes

Unzip the download. At **drive.google.com**, signed in to your **personal** account, click **New** → **Folder upload** and choose the **DPP Forms kit** folder. Wait until all 375 files are up. Keep the folder name. (Personal account, because your district blocks Apps Script on the school account.)

2 Make the template form · 3 minutes

At **forms.google.com** (personal account), start a **Blank form** and name it `DPP template`. Leave it with no questions. On its **Settings** tab:

- **Make this a quiz**: on. **Release grades**: Immediately after each submission. Respondents can see: **Missed questions** on, **Correct answers** off, **Point values** on.
- **Responses**: **Collect email addresses**: Verified. **Limit to 1 response**: on.

Every form the script builds is a copy of this one, so these settings carry over (correct answers stay hidden, so one period can't pass them to the next). Copy the template's **ID** from the address bar: the long code between `/d/` and `/edit`.

3 Make the script project · 5 minutes

a. Go to **script.google.com** (personal account) → **New project**. Click **Untitled project** and name it `DPP Forms builder`.
b. In **Code.gs**, select all the sample code and delete it. Open **Code.gs.txt** from the zip (it opens in Notepad, TextEdit or Text), select all, copy, and paste it into the editor. Click **Save** (the disk icon).
c. Click **Project Settings** (the gear on the left) and tick **Show "appsscript.json" manifest file in editor**. Go back to the **Editor** (the `< >` icon), click **appsscript.json**, select all, and paste the contents of **appsscript.json.txt** over it. Save.
d. Back in **Code.gs**, fill in three lines near the top, keeping the quote marks, then Save:

```
TEMPLATE_FORM_ID: '1AbC…xyz',             // the template's ID from step 2
SCHOOL_EMAIL: 'your school Google address', // the built folder is shared with it, view only
ONLY: 'C-NSO.1.2',                        // the test build in step 6; set back to '' in step 7
```

4 Turn on the Google Forms API · about 10 minutes

Apps Script can't set the answer key of a typed question or choose Verified emails, but Google's Forms API can. Using it takes a free Google Cloud project.

1. Go to **console.cloud.google.com**, signed in to your personal account. Agree to the terms if asked.
2. Click the project picker at the top → **New project**. Name: `DPP Forms`. Location: **No organization**. **Create**, then make sure **DPP Forms** is the project shown at the top.
3. ☰ **menu** → **APIs & Services** → **Library**. Search **Google Forms API** → **Enable**. Back in the Library, search **Google Drive API** → **Enable**. (Apps Script needs the Drive API turned on in a project like this one.)
4. ☰ **menu** → **Google Auth Platform** → **Get started**. App name: `DPP Forms`. User support email: your Gmail. **Next**. Audience: **External**. **Next**. Contact information: your Gmail. **Next**. Tick the agreement → **Continue** → **Create**.
5. **Google Auth Platform** → **Audience** → **Test users** → **Add users**: your Gmail → **Save**. Leave the publishing status on **Testing**.
6. ☰ **menu** → **Cloud overview** → **Dashboard**: copy the **Project number** (all digits, not the project ID).
7. In Apps Script: **Project Settings** → **Google Cloud Platform (GCP) Project** → **Change project**, paste the number, **Set project**.

> **Rather skip this step?** In Code.gs set `USE_FORMS_API: false`. The script builds everything else and lists the 166 typed answer keys in a **Typed keys to add** tab of the links sheet, one accepted answer per cell, with each form's edit link. You then add them by hand in each form (about an hour) and tick **Mark all other answers incorrect** on each.

5 Check the setup · 2 minutes

In the editor, choose **testSetup** in the function menu next to **Run**, and click **Run**. The first time, Google asks for permission: **Review permissions** → your personal account → **Google hasn't verified this app** → **Continue** (if you only see *Back to safety*, click **Advanced** → **Go to DPP Forms (unsafe)**). On the next screen tick **Select all**, then **Continue**. It's your own script; Google shows this warning for any script it hasn't reviewed.

The **Execution log** then lists **OK**, **NOTE** and **FIX** lines. Do what each FIX line says and run testSetup again until there are none.

6 Test build: one benchmark, 3 forms · 15 minutes

With `ONLY: 'C-NSO.1.2'`, choose **buildForms** and click **Run**. In a minute or two it builds that benchmark's Check, Re-check and Comparison, plus the links sheet: **My Drive** → **DPP Forms kit** → **DPP Forms (built)** → **DPP Forms links**. Then check one form all the way through, as a student would get it:

a. Signed in to your **school** account, open the links sheet (it's in **Shared with me**). On the **C-NSO.1.2 Check** row, click the **Copy link** → **Make a copy**.

b. In the copy, look over Settings: quiz on, correct answers hidden, Verified emails, 1 response. Click **Question 4** → **Answer key**. It should list `63 7/11`, `700/11` and `63 63/99`. **Note whether "Mark all other answers incorrect" is ticked.** The question should also still have its **Response validation** rule.

c. **Publish** the copy (leave Locked mode off for this test; it only works on school Chromebooks), open its link and take it: on Question 1 type `12 cm` (it should refuse it), then a wrong number such as `1/2`; on Question 4 type `700/11`. Submit and view your score: Question 1 should be marked wrong with 0 points, Question 4 right.

d. Done? In the copy, **Responses** → ⋮ → **Delete all responses**, then move the copy to the Trash.

> **If "Mark all other answers incorrect" is not ticked** (and Question 1 was left unmarked instead of scored 0): the keys are right, but Forms waits for you to grade any answer not on the key. Go ahead with the full build anyway. Then either tick that box once on each typed question in the built forms (166 questions, about 40 minutes), or tick it on the 1 to 5 typed questions each time you copy a form.

7 Full build · about an hour, on its own

Set `ONLY: ''` (two quote marks, nothing between), Save, choose **buildForms** and click **Run**. The first run works for about 5 minutes, then logs *"the next run starts by itself in about a minute"*. You can close the tab. Each run builds what it can in 5 minutes and starts the next, and the three test forms are kept. Watch the links sheet: each row's **Status** turns to **built**, and the last run logs *"All 102 forms are built"*. If the Status column stops changing for 15 minutes, run buildForms again; it picks up where it stopped.

> **Need some units first?** `ONLY` takes a list: `ONLY: 'Unit 9, Unit 10'` builds just those 12 forms (about 6 minutes). Set it back to `''` later and run buildForms again for the rest; forms already built are skipped.

Each week · about 2 minutes a form

1. Signed in to your **school** account, open the links sheet (bookmark it). Rows run in teaching order. Click the form's **Copy link** → **Make a copy**. The copy is in your school My Drive as "Copy of …"; students see only the form title.
2. If the sheet's **Locked mode** column says **on** (no-calculator forms), open the copy's **Settings** and turn on **Locked mode** under the quiz's Chromebook settings.
3. Click **Publish**, with responders limited to your school (not "Anyone with the link"), copy the link, and post it in your Google Classroom and John's.
4. **Responses** → **Link to Sheets**: link it to your one DPP responses spreadsheet, as in *00 Start here.pdf*. The rest of the week runs as that guide says.

If something goes wrong

What you see	What to do
FIX There is no folder named "DPP Forms kit"	Upload the folder (not the zip) to My Drive with that exact name (step 1).
Forms API 403: … isn't turned on	Step 4.3: enable the Google Forms API in the DPP Forms project. Wait a minute, then run again.
Turn on the Google Drive API …	Step 4.3: enable the Google Drive API too.
Access was refused, or "insufficient authentication scopes"	Run testSetup and tick **Select all** on the permission screen. Check the project number (step 4.7).
Google asks for permission again	Normal: a project in Testing asks again after 7 days. Click through as in step 5.

What you see	What to do
The Status column stopped changing	Run buildForms again. It skips built forms and rebuilds any half-built one. Google caps a personal account's background script time at 90 minutes a day, so on a slow day the last few forms may need a run the next day.
A row's Status says **failed: …**	Fix what it names (usually a missing picture: put it back in `images`), then run buildForms again. It rebuilds only those forms.
The copy link opens in your personal account	Switch to your school account in the top-right corner, or open the link in a browser window signed in only to your school account.
The copy link says you need access	In your personal Drive, share **DPP Forms (built)** with your school address as **Viewer** (SCHOOL_EMAIL does this). If it still refuses, the district is blocking files from personal accounts: ask your tech office to allow your own Gmail.
A spare form named like a check sits in My Drive	A run was cut off mid-form. Delete it; the real one is in DPP Forms (built) and in the links sheet.
You built the forms with an earlier copy of this kit	Keep them: the questions and keys are the same. Replace the `images` folder in **DPP Forms kit** with this kit's (the pictures now say "Question 1", "Question 2", …), paste this kit's `Code.gs` over the old code (copy your CONFIG lines across), Save, then run **updateImages** (about 10 minutes, continues by itself), **updateKeys** (typed answer keys, for example both the π-key and the 3.14 value) and **updateMessages** once. School copies you make afterwards get the new pictures; copies already made keep the old ones.
You want to start again from scratch	Run **startOver**. It moves every built form and the links sheet to the Trash (you can restore them for 30 days). Then run buildForms.

Good to know

- The forms in your personal Drive are masters. Edit one, and only copies made after that get the change.
- The script changes nothing outside the **DPP Forms kit** folder. It only reads the template, and testSetup makes one test form, which goes straight to the Trash.
- Typed answers accept every way the key lists (for example `63 7/11` or `700/11`). If a student types a correct answer the key missed, mark it correct under **Responses** → **Question**; Forms re-scores everyone who typed the same thing.
- The links sheet's **Edit link** opens the master; the **Copy link** makes a school copy. The **Form ID** column is for the script.